Input Parameters

Domain & Solver

Parameter	Case A (Decoupled)	Case B (Coupled)
Mesh	100×100	
Δx	$1\,\mu\mathrm{m}$	
Δt	$1 \times 10^{-6}\,\mathrm{s}$	$1.16 \times 10^{-10}\,\mathrm{s}$
l (interface width)	$4\,\mu\mathrm{m}$	
Loading	$U_y = \pm 0.1\,\mathrm{m\,s^{-1}}$	

Material 0 — Fe-Ni35 Grain

Parameter	Case A (Decoupled)	Case B (Coupled)
C_{11}	$1.40 \times 10^{11}\,\mathrm{Pa}$	
C_{12}	$9.20 \times 10^{10}\,\mathrm{Pa}$	
C_{44}	$1.11 \times 10^{11}\,\mathrm{Pa}$	
ρ_0	$7500\,\mathrm{kg\,m^{-3}}$	
N_{slips}	12 (FCC)	
CRSS	$1.6 \times 10^{7}\,\mathrm{Pa}$	
h_0	$1000\,\mathrm{MPa}$	
τ_s	$300\,\mathrm{MPa}$	
k	10	
Grain 0 orientation	$\phi_1 = 15^\circ$	
Grain 1 orientation	$\phi_1 = 0^\circ$	

Material 1 — Crack / Void

Parameter	Case A (Decoupled)	Case B (Coupled)
C_{11}	$0.1\,\mathrm{Pa}$	$1 \times 10^{9}\,\mathrm{Pa}$ (isotropic)
ρ_0	$6000\,\mathrm{kg\,m^{-3}}$	
N_{slips}	0	

Note: $C_{\mathrm{void}} = 0.1\,\mathrm{Pa}$ in Case A is stable because the stress is frozen in Stage 2 (no momentum solve). In Case B C_{void} is raised to $10^9\,\mathrm{Pa}$ and made isotropic ($C_{\mathrm{shear}} = C_{11}/3$) to limit the stiffness contrast at the void–grain interface.

Phase-Field Interface

Parameter	Case A (Decoupled)	Case B (Coupled)
$M^{\mathrm{gr\text{-}cr}}$	$10^{-3}\,\mathrm{m^3\,J^{-1}\,s^{-1}}$	
$E^{\mathrm{gr\text{-}cr}}$	$0.5\,\mathrm{J\,m^{-2}}$	$0.1\,\mathrm{J\,m^{-2}}$